Analytical Synopsis: Comparative Statistical Analysis of Visa Inc. (Growth) and Coca-Cola (Value)

Executive Summary: This report applies statistical methods to evaluate the weekly performance of Visa Inc. (V) and The Coca-Cola Company (KO) from March 2025 to March 2026. By analyzing descriptive statistics, risk-adjusted returns, and linear regression, we distinguish the performance characteristics of a high-capitalization growth stock (Visa) versus a classic value stock (Coca-Cola).

1. Descriptive Statistics: Risk and Volatility

In *Business Statistics*, the **Standard Deviation** is the primary measure of investment risk. Our data confirms the theoretical profiles of these assets:

- **Visa (Growth Profile):** Visa exhibited a higher standard deviation of **0.0263**. As a growth stock, its value is tied to future earnings potential and technology adoption, leading to higher price swings. This is evidenced by a wide **Range of 15.14%**.
- **Coca-Cola (Value Profile):** Coca-Cola showed a lower standard deviation of **0.0195** and a narrower **Range of 11.18%**. This reflects the stability of a value stock—an established company with predictable cash flows and lower sensitivity to market euphoria.

2. Risk-Adjusted Performance (Sharpe Ratio)

A key concept in financial statistics is the **Sharpe Ratio**, which measures the "excess return" per unit of risk.

- Using a weekly risk-free rate of **0.038%**, both companies yielded negative Sharpe Ratios (Visa: **-0.025**; Coca-Cola: **-0.039**).
- While both underperformed the risk-free rate during this specific period, Visa's slightly higher ratio suggests that for the risk taken, the growth-oriented "upside" potential provided marginally better compensation than the stagnating value of Coca-Cola.

3. Regression Analysis: Market Relationship

We performed a linear regression to understand the **Coefficient of Determination (R^2)**, which communicates how much of one variable's movement explains another.

- **The Relationship:** The **Multiple R** (Correlation Coefficient) is **0.322**, indicating a weak positive relationship.
- **The R^2 Value:** The R^2 is **0.1036**, meaning only about **10.4%** of the variation in Visa's returns can be explained by Coca-Cola's performance. This statistical independence is vital for portfolio diversification; it proves that the growth drivers of a fintech leader (Visa) are fundamentally different from the consumer staple drivers of Coca-Cola.

4. Conclusion

The data supports the textbook definitions of these asset classes. **Coca-Cola** functions as a low-volatility **Value Stock**, providing stability during market fluctuations. **Visa** operates as a **Growth Stock**, characterized by higher variance (risk) but offering a different return profile that is not heavily dictated by the movements of traditional value assets. For an investor, the low R^2 suggests that holding both stocks would effectively reduce idiosyncratic risk within a portfolio.